1 Introduction

Lecture 9 is about polar codes.

2 Review: A General Channel Q

x represents the input to the channel and y represents ... and I missed it.

2.1 Coding Theory

Applications of information theory to electrical engineering:

- Data compression: Given data, we want to represent/encode it using a smaller number of bits
- Error correcting codes: Communicating messages over noisy channels. Storage of information

The main problem of the codes that we described so far is that they are not explicit and hence encoding and decoding is inefficient.

Coding theory: The design of explicit codes.

2.2 Linear Codes

$$a \to f(a) \to Q^{\otimes}(f(a)) \to g(Q^{\otimes}(f(a)))$$

An error correcting code is linear if the encoding function

$$f : \{0,1\}^m \to \{0,1\}^n$$

is linear (mod 2). f can be described by a matrix. A random linear transformation is a good error correcting code. There are many explicit constructions.

2.3 What this lecture is about

- Simple, elegant, linear, useful, cool
- Reach channel capacity
- Proof uses information theory; a nice application

3 The Ants Puzzle

At times $n = 0$, one ant on point $s \in [0,1]$. In every time-step $n = 1, 2, 3, \ldots$, every any on the interval duplicates into two ants. If the original ant stood on point $t \in [0,1]$, let $u = t(1-t)$. Then, one of the two ants moves to point $t + u = 2t - t^2$ and the other moves to point $t - u = t^2$.

3.1 Prove

Prove that for every δ, there exists n such that at time n at most a fraction of δ of the population of ants is in the interval $[\delta, 1-\delta]$. That is, a fraction of at least $1-\delta$ is in $[0,\delta] \cup [1-\delta, 1]$ (most ants are close to one of the poles).

3.1.1 Hint

You have to analyze the variance of the population, of the distribution. What happens, when one ant splits to two, the variance of the location increased, but the expectation remained the same. The average remained in the same point, so the expectation of the location remains s, but the variance increases until it reaches the poles.

4 Erasure Channel

Input is $x \in \{0,1\}$ and the output is $y \in \{0,1,*\}$. Let $0 < \epsilon < 1$. A bit is "erased" (replaced by $*$) with probability ϵ.

- If alice sends 0, Bob gets 0 with probability $1-\epsilon$ and $*$ with probability ϵ.
- If alice sends 1, Bob gets 1 with probability $1-\epsilon$ and $*$ with probability ϵ.

4.1 Capacity of Erasure Channel

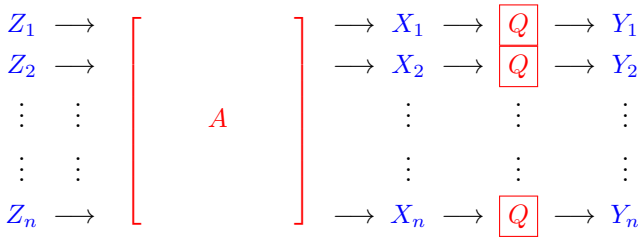
If $X \sim (p, 1-p)$ then $Y \sim (p \cdot (1-\epsilon), \epsilon, (1-p) \cdot (1-\epsilon))$

$$\begin{aligned} H(Y \mid X) &= H(\epsilon, (1-\epsilon)) \\ H(Y) &= H(\epsilon, (1-\epsilon)) + (1-\epsilon) \cdot H(p, (1-p)) \\ I(Y;X) &= H(Y) - H(Y \mid X) = (1-\epsilon) \cdot H(p, (1-p)) \\ c_Q &= \max_{P_X} I(Y;X) = 1-\epsilon \end{aligned}$$

There are two trivial cases:

- If $\epsilon = 0$ then $c_Q = 1$
- If $\epsilon = 1$ then $c_Q = 0$

4.2 A nice picture



There are n copies of the channel Q.

$A : \{0,1\}^n \to \{0,1\}^n$ is an invertible linear transformation (one to one). Here $A(Z) = X$, $A^{-1}(X) = Z$, $X \in_R \{0,1\}^n$ uniform, and $Z \in_R \{0,1\}^n$ uniform.

First, we know $I(Z;Y) = I(X;Y) = n \cdot c_Q$ (since A is invertible). Now, via the chain rules, we can write

$$I(Z;Y) = \sum_{i=1}^{n} I(Z_i; Y \mid Z_{<i})$$

4.2.1 Comment

Since A is linear and Q symmetric, $I(Z_i; Y \mid Z_{<i} = b)$ doesn't depend on b. Changing b just adds an additive constant in $\{0,1\}$ to each variable $X_1, \ldots, X_n$ and hence is just a change of names $0, 1$ at the input to the channel. This doesn't change the mutual information because the channel is symmetric.

4.2.2 Assumption

A is such that for every i, $I(Z_i; Y \mid Z_{<i})$ is either 0 or 1. For $i \in S$, $I(Z_i; Y \mid Z_{<i}) = 0$, for $i \notin S$, $I(Z_i; Y \mid Z_{<i}) = 1$

4.2.3 Encoding

For $i \in S$, fix $Z_i = 0$. For $i \notin S$, fix Z_i to be a message bit (we assume for simplicity a uniformly random message).

4.2.4 Decoding

Given Y, inductively over i, we figure out Z. Assume by induction that we already know $Z_{<i}$. If $i \in S$, $Z_i = 0$. If $i \notin S$, $I(Z_i; Y \mid Z_{<i}) = 1$, so Z_i is determined by $Z_{<i}, Y$. Done! But why would we ever have such an amazing property?

Observation: It is enough to have this property approximately

4.2.5 Approximately

We can make the following changes:

1. A is such that for almost every i, $I(Z_i; Y \mid Z_{<i})$ is either very close to 0 or very close to 1.
2. For $i \in S$, $I(Z_i; Y \mid Z_{<i}) \leq \delta$, for $i \in T$, $I(Z_i; Y \mid Z_{<i}) \geq 1 - \delta$
3. For $i \notin T$, fix $Z_i = 0$.
4. For $i \in T$, fix Z_i to be a message bit (we assume for simplicity a uniformly random message).
5. If $i \notin T$, $Z_i = 0$. If $i \in T$, $I(Z_i; Y \mid Z_{<i}) \geq 1 - \delta$, so Z_i is almost determined by $Z_{<i}$, Y.

Observation: It is enough to have this property approximately!

4.2.6 Another comment

Since Z_i is independent of $Z_{<i}$

$$\begin{aligned} I(Z_i; Y \mid Z_{<i}) &= H(Z_i \mid Z_{<i}) - H(Z_i \mid Y, Z_{<i}) \\ &= H(Z_i) - H(Z_i \mid Y, Z_{<i}) \\ &= I(Z_i; Y, Z_{<i}) \end{aligned}$$

so we can write

$$I(Z; Y) = I(X; Y) = n \cdot c_Q = \sum_{i=1}^{n} I(Z_i; Y \mid Z_{<i}) = \sum_{i=1}^{n} I(Z_i; Y, Z_{<i})$$

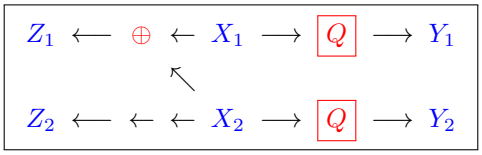
Goal: A such that for almost every i, $I(Z_i; Y, Z_{<i})$ is very close to 0 or 1.
Comment: We can view $I(Z_i; Y, Z_{<i})$ as the capacity of the channel

$$Z_i \to Y, Z_{<i}$$

Conceptually, we replaced n original channels by n "virtual channels"

4.3 How do we get such a matrix A

Consider the simplest 2×2 matrix A^{-1}



$Z_2 = X_2$ and $Z_1 = X_1 \oplus X_2$, which is equivalently $X_2 = Z_2$ and $X_1 = Z_1 \oplus Z_2$.

First, what is $I(Z_1; Y)$? If we know Y we know Z_1 if and only if $Y_1 \neq *$ **and** $Y_2 \neq *$ (probability $(1-\epsilon)^2$). Otherwise, Z_1 is uniform. Thus $I(Z_1; Y) = (1-\epsilon)^2$. The channel $Z_1 \to Y$ is an eraser channel with erasure probability $2\epsilon - \epsilon^2$

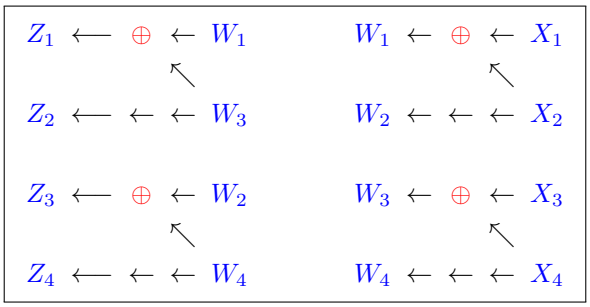
Second, what is $I(Z_2; Y, Z_1)$? If we know Y, Z_1 we know Z_2 if and only if $Y_1 \neq *$ **or** $Y_2 \neq *$ (probability $1 - \epsilon^2$). Otherwise, Z_1 is uniform. Thus $I(Z_2; Y, Z_1) = 1 - \epsilon^2$. $Z_2 \to Y, Z_1$ is an eraser channel with erasure probability ϵ^2

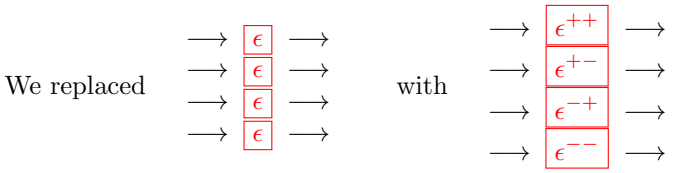
4.3.1 Now what do we get?

We replace the original setup with two virtual channels:

1. $Z_1 \to Y$ is an eraser channel with erasure probability $\epsilon^+ = 2\epsilon - \epsilon^2$
2. $Z_2 \to Y, Z_1$ is an eraser channel with erasure probability $\epsilon^- = \epsilon^2$

where $\epsilon^+ + \epsilon^- = 2\epsilon$. This is local polarization. This should remind us of our ant puzzle.





Recursion: Increase n by a factor of 2 in each step. Recursive construction on each of two blocks of size $n/2$. Match each channel to the corresponding channel in the other block. Apply one more step. After $\log n$ steps we get our final n channels. By the ants puzzle, for almost every i, $I(Z_i; Y, Z_{<i})$ is very close to 0 or very close to 1.

4.3.2 Final Result

For almost every i, $I(Z_i; Y, Z_{<i})$ is very close to 0 or very close to 1.

- For $\approx c_Q \cdot n$ coordinates i, $I(Z_i; Y, Z_{<i}) > 1 - \delta$
- For $\approx (1 - c_Q) \cdot n$ coordinates i, $I(Z_i; Y, Z_{<i}) < \delta$

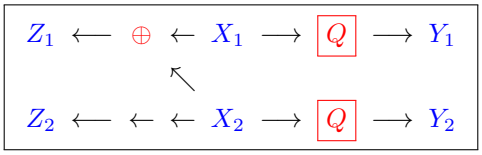
By analyzing the variance

$$\delta \approx 2^{-\sqrt{n}}$$

which is exponentially small!

5 General Channel (Symmetric, Binary Input)

The same construction:



where $Z_2 = X_2$ and $Z_1 = X_1 \oplus X_2$, or equivalently $X_2 = Z_2$ and $X_1 = Z_1 \oplus Z_2$. We know $I(X_1; Y) = I(X_2; Y) = c_Q$. Hence

$$I(Z_1; Y) + I(Z_2; Y \mid Z_1) = 2c_Q$$

This is again a local polarization!

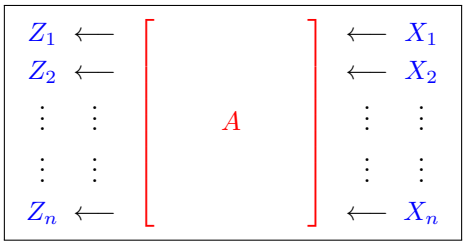
Usually, the first is small, as, since $Z_1 = X_1 \oplus X_2$, it gets the noise from both channels, and the second is large, as, since it's conditioned on knowing Z_1, it obtains information from both channels.

This givens an excellent code for many channels!

5.1 Example: BSC(p)

- $I(Z_1; Y) = I(X_1 \oplus X_2; Y_1 \oplus Y_2)$ equivalent to BSC($2p - p^2$)
- $I(Z_2; Y \mid Z_1) = I(X_2; Y \mid X_1 \oplus X_2)$, equivalent to the information obtained by sending X_2 twice, over BSC(p)

6 Polar Codes for Compression



$A(Z) = X$, $A^{-1}(X) = Z$, $X_1, \ldots, X_n$ are i.i.d., and $X_i \sim (p, 1-p)$. Now we have

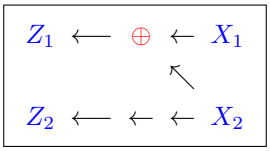
- $H(Z) = H(X) = n \cdot H(p, 1-p)$

- $H(Z) = \sum_i H(Z_i \mid Z_{<i})$

We want $H(Z_i \mid Z_{<i})$ to be close to 0 or 1.

We take the set of Z_i-s with $H(Z_i \mid Z_{<i})$ close to 1 (other Z_i-s don't give much information about X)

6.1 The setup for two channels



... there is more in a picture.

7 Error Correcting Codes in Discrete Math

An error correcting code is a function $f : \{0,1\}^m \to \{0,1\}^n$, such that, for ever $x \neq x' \in \{0,1\}^m$, we have

$$|f(x) \oplus f(x')| \geq k$$

There are many explicit constructions in all range of parameters.

Error correcting codes can be used to communicate over adversarial channels. ... and I missed the rest.

8 Communication Complexity of Equality

Alice has $x \in \{0,1\}^m$. Bob has $x' \in \{0,1\}^m$. They want to check if $x = x'$, with minimum communication.

Let $f : \{0,1\}^m \to \{0,1\}^n$ be an error correcting code with $k \geq \frac{n}{4}$ (agreed upon beforehand). Alice chooses a random $i \in [n]$ and sends to Bob i, $f(x)_i$. Bob checks if $f(x)_i = f(x')_i$.

- If $x = x'$, then $P[f(x)_i = f(x')_i] = 1$
- If $x \neq x'$, then $P[f(x)_i = f(x')_i] = \frac{3}{4}$

Repeat this protocol 100 times (possibly in parallel), to get confidence, if in one of these times they get $f(x)_i \neq f(x')_i$, they know that $x \neq x'$, otherwise, $x = x'$ is very likely.

9 7-Hat Problem

$$(X_1, X_2, X_3, X_4, X_5, X_6, X_7) \in_R \{0,1\}^7$$

There are 7 players where player i has bit X_i on their forehead. Can see all other bits, but not X_i. Simultaneously, each player can guess or not guess their

bit. They all win if at least one player guessed correctly and no player guessed incorrectly. They can plan in advance, but no communication after seeing the bits.

What is their maximal success rate? I would guess $\frac{1}{2}$. But it is wrong!

9.1 Protocol for three players

A player that sees two different bits (01 or 00) doesn't guess. A player that sees two equal bits (00 or 11) guesses the opposite bit.

If two bits are the same and one is different (001, 010, 100, 011, 101. 110), only the play that has the different bit guesses, and that guess is correct.

If all bits are equal (000, 111), all players guess, incorrectly.

This results in a winning probability of $\frac{3}{4}$.

9.2 Hamming Code

We need a function $f : \{0,1\}^4 \to \{0,1\}^7$ such that

$$f(Y_1, Y_2, Y_3, Y_4) = (Y_1, Y_2, Y_3, Y_4, Y_1 \oplus Y_2 \oplus Y_4, Y_1 \oplus Y_3 \oplus Y_4, Y_2 \oplus Y_3 \oplus Y_4)$$

Any element in $\{0,1\}^7$ is at distance 0 or 1 to a codeword.

9.2.1 Protocol for 7 players

A player that sees 6 bits that are not consistent with any codeword, doesn't guess. A play that sees 6 bits that are consistent with a codeword, guesses that bit that is not consistent with that codeword.

If 6 out of 7 bits are consistent with a codeword and the last one is inconsistent with that codeword, only the player that has the inconsistent bit guesses, and that guess is correct.

If all bits are consistent with a codeword, all players guess, incorrectly. The winning probability is $\frac{7}{8}$.